RARES IACOB

SOFTWARE DEVELOPER | JAVA / SPRING BOOT BACKEND

Timișoara, Romania | +40 754 650 236 | raresiacob26@gmail.com
github.com/raresiacob3 | linkedin.com/in/rares-iacob-755380230

PROFILE

Software developer with automotive Embedded C experience and a background in production troubleshooting, focused on Java/Spring Boot backend development. Built and published ScentVault (iOS and Android) with AI-based image recognition, cloud APIs and subscription billing, and built JobPilot, a multi-user Spring Boot/PostgreSQL application with rotating JWT refresh tokens and versioned Flyway migrations. Adding Python/FastAPI and LLM integration to my backend toolkit.

TECHNICAL SKILLS

Backend: Java 17/21, Spring Boot 3, Spring Security, JPA/Hibernate, REST APIs, Python, FastAPI, Node.js/Express

Data: PostgreSQL, SQL, Flyway, Alembic, MongoDB, Firebase Auth/Firestore

Frontend & mobile: React, TypeScript, React Native/Expo

DevOps & testing: Docker, Docker Compose, GitHub Actions, Google Cloud Run, JUnit, Mockito, pytest, Git, OpenAPI

AI: OpenAI API (text, vision), BM25 document retrieval, retrieval evaluation (recall@k, MRR), LLM explanations with citations; ComfyUI and LoRA training (experiments)

Embedded: Embedded C, AUTOSAR, VectorCAST, Google Test, CANoe, DaVinci, Wireshark

SELECTED PROJECTS

JobPilot | Full-Stack Job Application Tracker github.com/raresiacob3/jobpilot

Java 21 • Spring Boot 3 • Spring Security • PostgreSQL • Flyway • React • TypeScript • Docker

- Built a multi-user REST API with JWT access tokens and server-side refresh token rotation with reuse detection, access roles and strict per user data isolation (another user's resources return 404).
- Enforced the application pipeline as a server-side state machine (invalid transitions return 409) and recorded activity events in the same transaction as each change.
- Versioned the schema with Flyway (Hibernate validate mode), used grouped aggregate queries instead of N+1 for analytics, and shipped multi-stage Docker builds, Docker Compose and OpenAPI docs.
- Wrote unit tests (JUnit, Mockito) for the status state machine, service rules and JWT handling.

ScentVault | Published AI-Powered Perfume Tracking App App Store + Google Play

React Native • Expo • TypeScript • Node.js • Firebase • OpenAI API • Google Cloud Run • RevenueCat

- Built and released a mobile product on iOS and Android for fragrance recognition, collections, wishlists and statistics.
- Integrated AI-based image recognition returning structured matches with confidence signals, Firebase authentication, a cloud-hosted backend and RevenueCat subscription billing.
- Managed EAS production builds, app store submissions and review cycles.

Supplier Intelligence | Python Backend & Applied AI github.com/raresiacob3/supplier-intelligence

Python • FastAPI • SQLAlchemy • PostgreSQL • Alembic • Pydantic • pytest • Docker • OpenAI Responses API

- Built a FastAPI/PostgreSQL service that ranks suppliers with a deterministic weighted score computed in a single SQL query, with a per-criterion breakdown; CSV ingestion validates every row with Pydantic and runs off the async event loop.
- Implemented BM25 retrieval over chunked supplier documents feeding an optional LLM explanation endpoint with document citations; measured retrieval offline on a golden set (recall@1 0.9, MRR 0.9), enforced as a regression test.
- Versioned the schema with Alembic and added request-ID logging; 42 pytest tests (about 94% coverage) run in GitHub Actions on SQLite and PostgreSQL, plus a Docker Compose smoke test. AI-assisted, synthetic data, not deployed.

Additional projects

- **MemoryVials** (Java 17, Spring Boot, MongoDB): perfume e-commerce backend with catalog, cart, stock checks, checkout and order tracking.
- **Questify** (Java, Spring Boot, React): gamified task manager with quests, XP, difficulty levels and filtering.
- **Generative AI experiments** (ComfyUI, FLUX.2 Klein, SDXL, SimpleTuner, RunPod): trained a custom LoRA on a cloud GPU and built diffusion workflows.

PROFESSIONAL EXPERIENCE

Technical Consultant | The Access Group Jan 2024 – Present

Previously Associate Technical Engineer

- Investigate complex production incidents in ConQuest and EasyBuild: application defects, data integrity, file corruption, licensing and client environment configuration.
- Use SQL, application logs and controlled database operations to isolate root causes and validate data.
- Reproduce defects and document findings for second-line support and engineering/QA teams; deliver remote sessions and technical explanations to customers.

Junior Software Developer | Magna Electronics Dec 2022 – Jan 2024

- Developed Embedded C functionality for the Imaging22 automotive radar sensor platform, including a board heater component from detailed design through implementation and unit testing.
- Authored Detailed Design Documents, took part in engineering reviews and project audits, and documented implementation and verification decisions.
- Designed and ran unit tests with Google Test and VectorCAST; investigated failures and debugged in Visual Studio.
- Automated engineering tasks with Python scripts and contributed to internal REST APIs and tooling; worked with Git, Jira and CI/CD.

AUTOSAR Intern | FEV ECE Jul 2022 – Dec 2022

- Implemented and updated AUTOSAR application modules in Embedded C with CANoe and DaVinci.
- Built Python test and engineering utilities and analyzed Ethernet traffic with Wireshark.

EDUCATION

B.Sc. Computer Engineering | Politehnica University Timișoara 2020 – 2024

ADDITIONAL

Languages: Romanian (native), English (advanced)

Volunteering: Intel Liga AC Labs; Developer Students Club; Timișoara 2023 – European Capital of Culture